

Introduction to Statistics

What is Statistics?

Statistics is **learning** from **data**

Statistics

Five Stages of Statistical procedure-

1. Data Collection
2. Organization
3. Presentation
4. Analysis
5. Interpretation/ conclusion

Statistics

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Statistics refers to the scientific methods for **collecting**, **organizing**, summarizing, **presenting**, and **analyzing** data, and drawing a **valid conclusion**.

Example: Child malnutrition status, Monthly expenditure of citizens of a city, Relationship of crime with space and time, Number of active users in a day of a website, average lifetime of the people of a country etc.

Population and Sample in Statistics

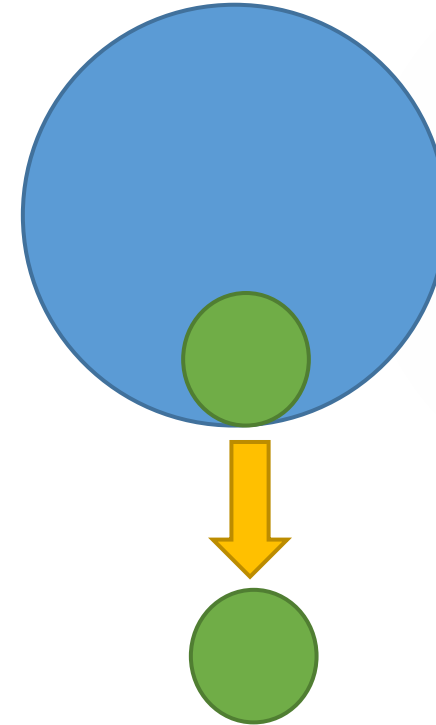
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Population:

Population is the **collection of all items** or individuals of interest in a particular study

Sample:

A representative **part of the population** of interest

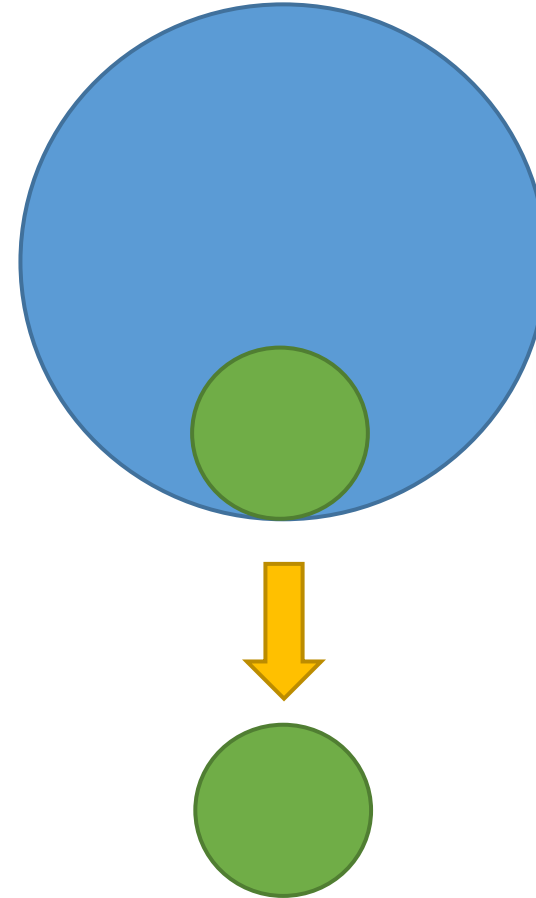


Population and Sample in Statistics

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Population: All citizens of Dhaka
Size: 2 crore

Sample: Some citizens of Dhaka
Size: 2000



•**Population:** All websites on the internet that experience cybersecurity attacks.

•**Sample:** A subset of 500 randomly selected websites analyzed for cybersecurity vulnerabilities.

Another example:

•**Population:** All software engineers working in a multinational tech company.

•**Sample:** A group of 200 software engineers.

Parameter & Statistic

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Parameter:

A **constant** which is a **function of population values**, and is **usually unknown**, is called a parameter

Number of smokers (population) = 1 crore
i.e., 50%

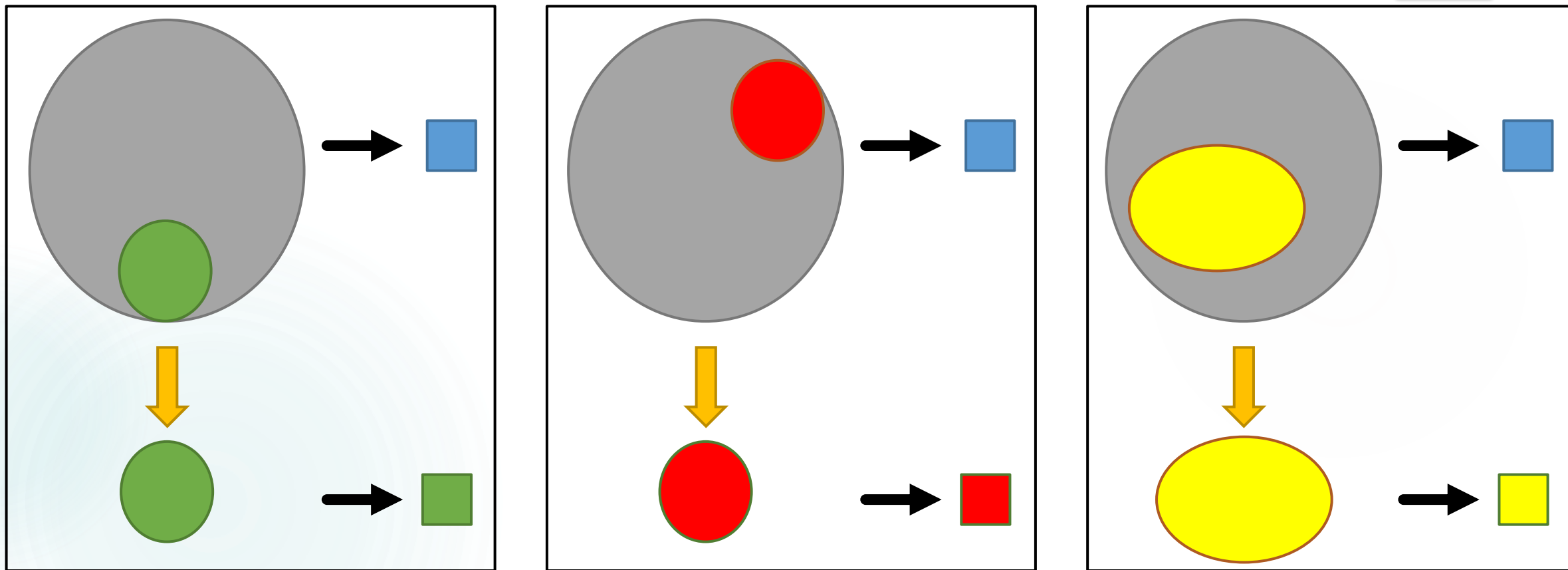
Statistic:

Any **function of sample values**, which is an **estimate of the parameter** and which is a **known** value, is called a statistic

Number of smokers (sample) = 1200
i.e., 52%

Parameter & Statistic

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Parameter



Statistic 1



Statistic 2



Statistic 3

•**Parameter (Population Measure):** The true average response time of all servers in a data center (e.g., the actual mean response time of **all** servers is 120 ms).

•**Statistic (Sample Measure):** The average response time of **a randomly selected** 50 servers from the data center (e.g., the sample mean response time is 118 ms).

Another Example:

•**Parameter:** The actual proportion of all software engineers in a company who prefer Python over Java.

•**Statistic:** The proportion of **a surveyed group** of 100 software engineers who prefer Python over Java.

Types of Statistics

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- ❑ Descriptive Statistics
- ❑ Inferential Statistics

Types of Statistics

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- Descriptive Statistics

Methods for **organizing**, **summarizing** and **presenting** data in an informative way

- Inferential Statistics

Methods for **estimating a parameter** of a population on the **basis of a sample**

Scopes of Statistics

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- ❑ State and administration
- ❑ Medical science
- ❑ Social Sciences
- ❑ Economics
- ❑ Artificial Intelligence
- ❑ Demography
- ❑ Agriculture
- ❑ Business and management
- ❑ Research etc.

Variable

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Case No.	Gender	Age	Smoking status
Person 1	Gender (Person 1) = Male	Age (person 1) = 40	Smoker (Person 1) = Yes
Person 2	Gender (Person 2) = Male	Age (person 1) = 20	Smoker (Person 1) = No
Person 3	Gender (Person 3) = Female	Age (person 1) = 32	Smoker (Person 1) = No
Person 4	Gender (Person 4) = Male	Age (person 1) = 59	Smoker (Person 1) = No
Person 5	Gender (Person 5) = Female	Age (person 1) = 38	Smoker (Person 1) = Yes

Variable

- ▶ A variable is a **characteristic**, containing two or more values or categories that can **vary** from person to person, object to object, or phenomenon to phenomenon
- ▶ Example: Gender, Age, Educational status, Hair color, religion, Place of residence, Monthly income, Satisfaction level, Soap brand, Temperature, GPA etc.

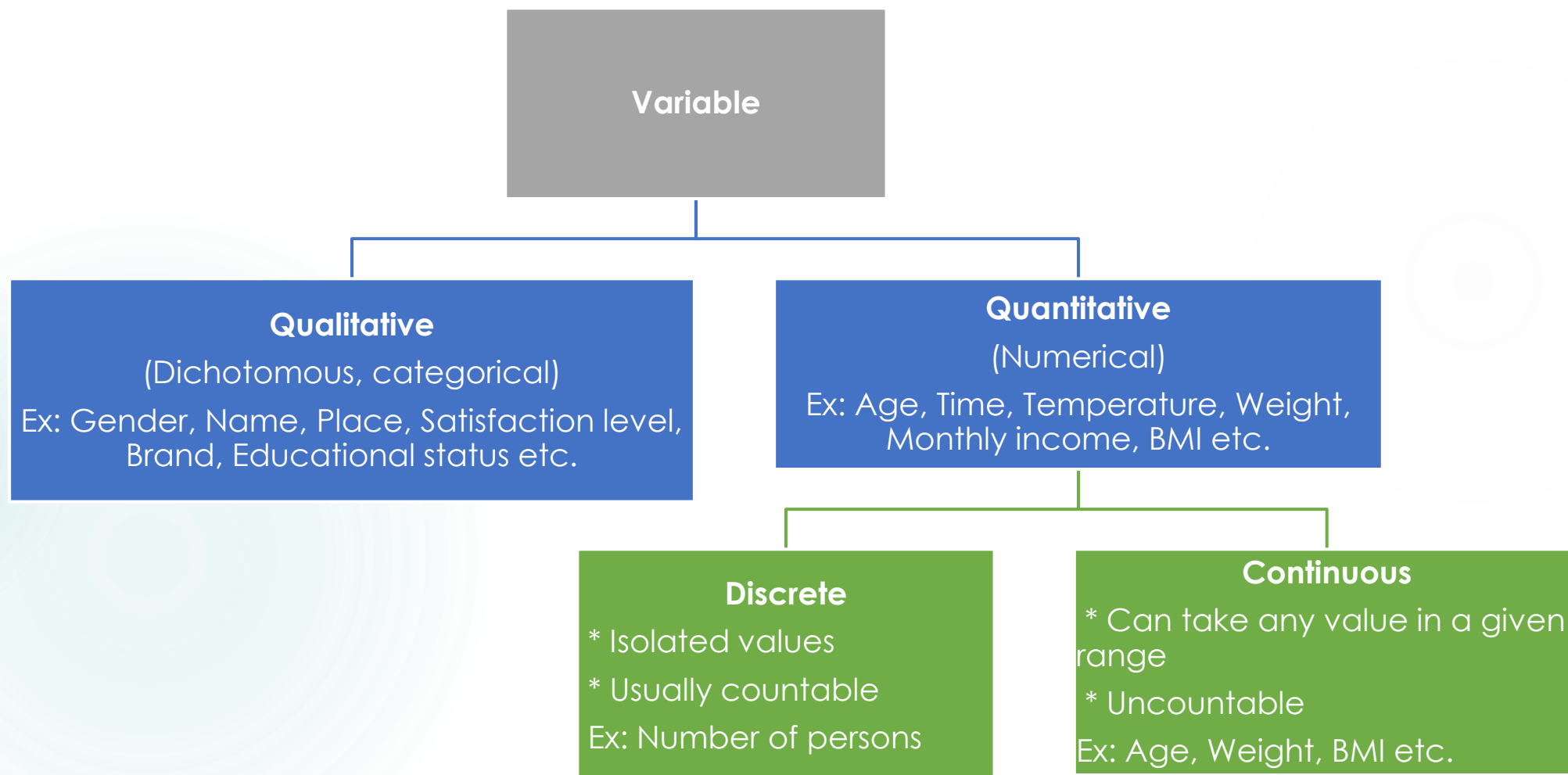
Variable

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Variable	Possible Values / Categories
Gender	Male, Female
Age	10, 50, 36, 18, 29, 75 etc.
Highest education level	Primary, Secondary, Higher etc.
Number of employees	10, 50, 89, 125, 4562 etc.
Salary	\$1000, \$10000, IR45000, BDT 98000, Rs.500000 etc.
Duration	10 hours, 2 days, 4 weeks, 10 years etc.
Weight	45 lb., 60 kg, 900 gm, 5 ton etc.
Wealth status	Poor, middle, Higher etc.

Classification of Variables

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Classification of Variables

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Class Task:

Find the types of the following variables-

- Monthly salary:
- Soap Brand:
- Occupation:
- Color:
- Weight:
- Duration of a class:
- Number of family members:
- Satisfaction level:
- Religion:
- Temperature:
- Food flavor:
- Wealth Status:
- Highest education level:
- Nationality:

Level of measurement

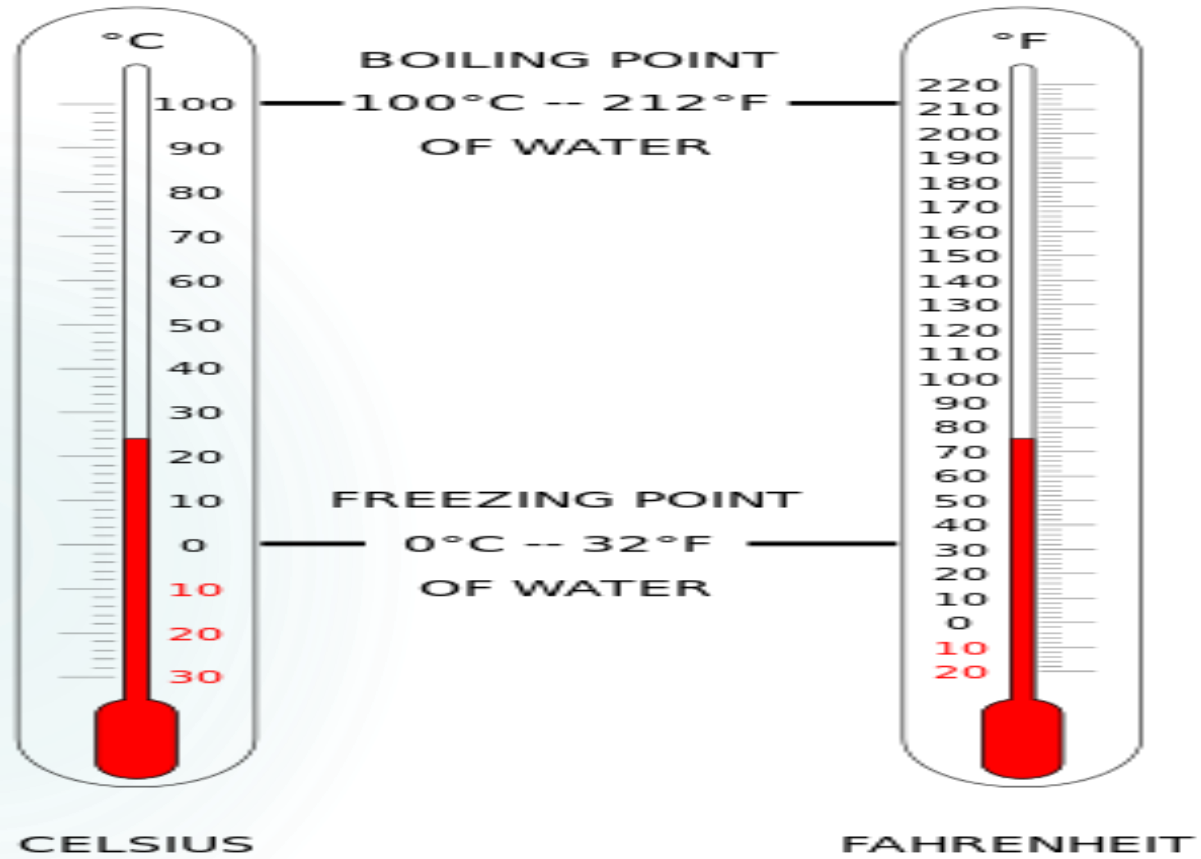
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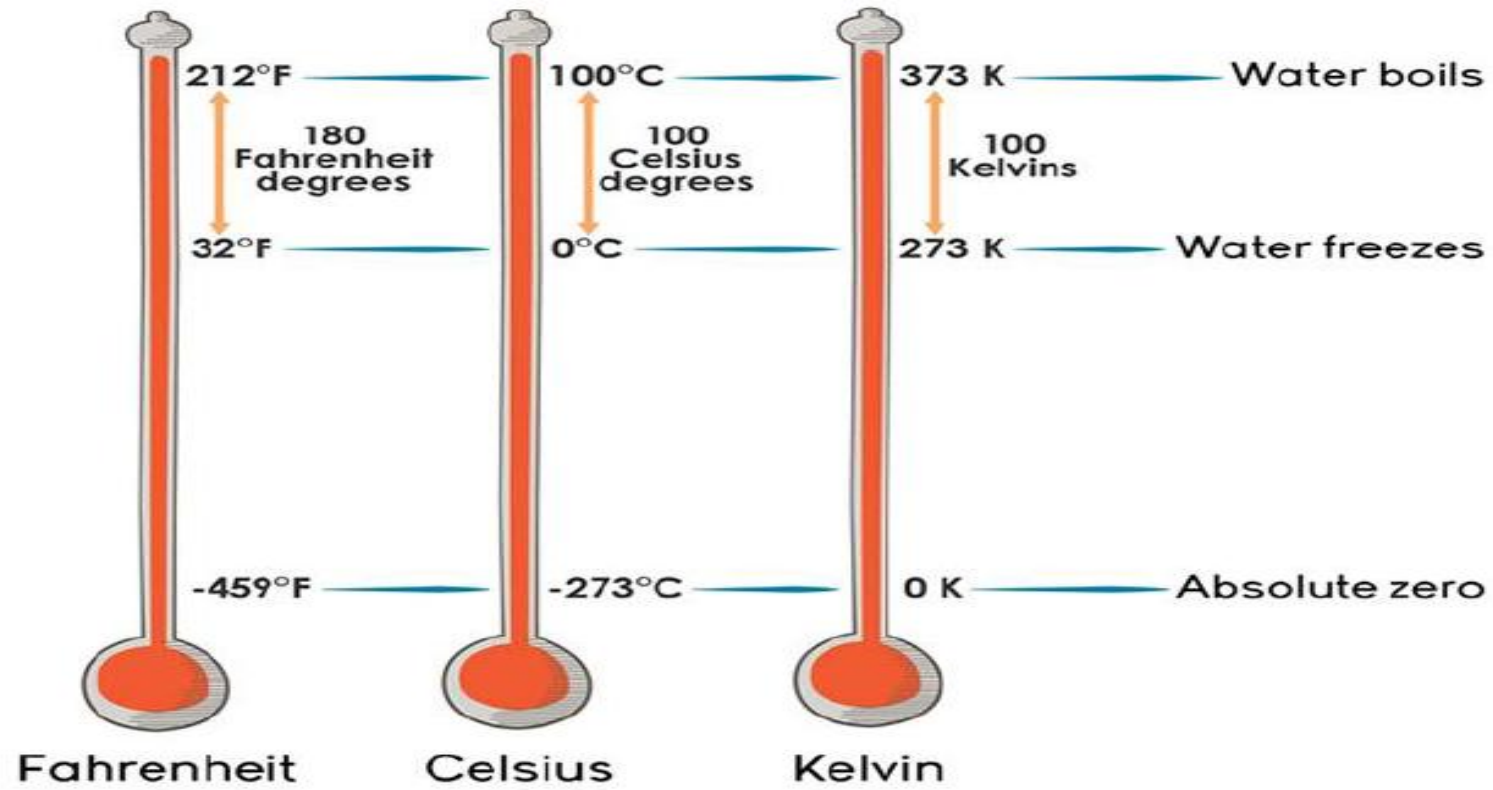
Variable type	Scale of measurement
Qualitative	Nominal: • Names or categories • Cannot be ordered or numerically measured <u>For example:</u> person's name, gender, place of resident, brand name etc.
	Ordinal: • Categories • Can be ordered • Cannot be numerically measured <u>For example:</u> Wealth status: poor, middle, rich; Education: primary, secondary, higher etc.

Level of measurement

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Variable type	Scale of measurement
Quantitative	Interval: • Numerically measured • Can be find differences, but not ratios • Does not have true or meaningful zero point <u>For example:</u> Temperature: 0°C temperature does not mean that there is no heat. It will read 32° in Fahrenheit scale!
	Ratio: • Numerically measured • Can be find differences and also ratios • Does have true or meaningful zero point <u>For example:</u> Height: 0 cm height means 'no' height, Distance: 0 meter distance means 'no' distance etc.





Level of measurement

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Class Task:

Find the level of measurement of the following variables-

- Monthly salary:
- Soap Brand:
- Occupation:
- Color:
- Weight:
- Duration of a class:
- Number of family members:
- Satisfaction level:
- Religion:
- Temperature:
- Food flavor:
- Wealth Status:
- Highest education level:
- Nationality
- IQ level

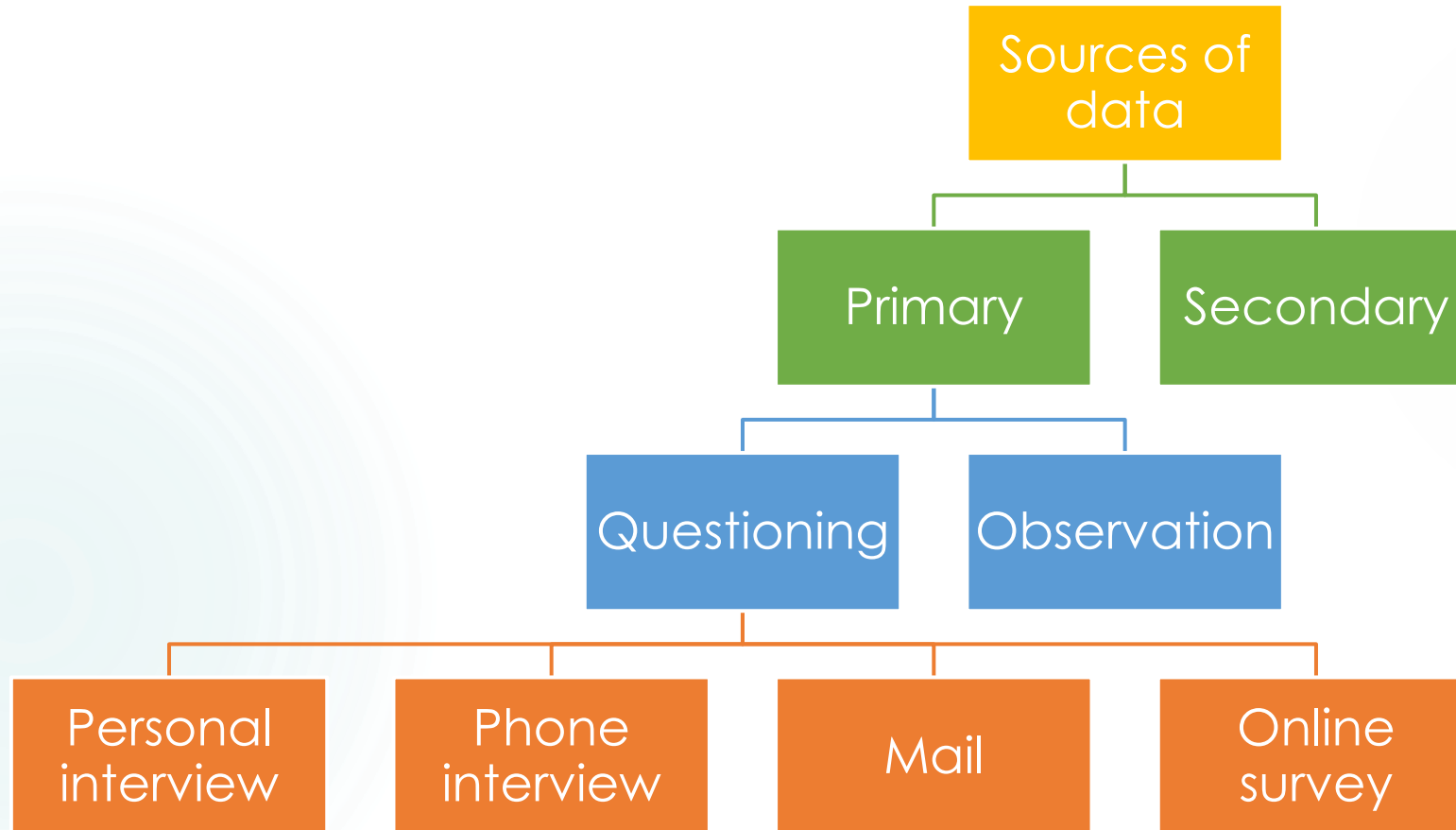
Data

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Data are raw, disorganized facts and figures collected from any field of inquiry.

Sources of data

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Sources of data

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